

Coding Scheme to Answer Each RQ

RQ1: IoT sensing and architecture

“What IoT architecture and sensing technologies are most effective for reliable bin fill-level monitoring in smart waste management systems?”

In your coding scheme, you extract:

- Sensor type (ultrasonic, ToF, etc.)
- Sensor placement (lid, side, inside)
- Communication method (Wi-Fi, etc.)
- Power source (battery, solar)
- Sampling frequency
- Calibration & ground truth method
- Any reported accuracy issues or environmental challenges

How it helps answer RQ1:

By comparing these fields across papers, you can see:

- Which sensors are used most frequently
- Which architectures produce the most reliable readings
- What problems occur (angle, temperature, etc.)
- What best practices the literature agrees on

RQ2: Forecasting models and accuracy

“Which machine learning or time-series models have achieved the highest accuracy in predicting bin fill levels or waste generation trends?”

In the coding scheme, extract:

- Type of forecasting model (LSTM, XGBoost, CNN, GNN, etc.)
- Features used (lag values, weather, day-of-week, events, bin type)
- Prediction horizon (1h, 6h, 1 day...)
- Dataset size
- Evaluation metrics (MAE, RMSE, MAPE, R^2)
- Whether uncertainty/quantiles (predication range for service window routing) were used

How it helps answer RQ2:

You can compare models based on:

- Which ones achieved the lowest error
 - Which performed poorly and why
 - What types of data and features lead to high accuracy
 - Whether deep learning actually outperformed classical models
-

RQ3: Optimization and routing with forecasts

“What optimization methods are most effective for dynamic waste collection routing, and how are forecasting results integrated into route planning?”

Extract:

- Optimization type (OR-Tools, heuristics, Reinforcement Learning, etc.)
- Constraints (truck capacity, time windows, distance, travel time)
- Objective function (minimize distance, minimize CO₂ , minimize bin overflow, etc.)
- Integration method:
 - Static routing
 - Threshold-based
 - Predict-then-optimize
- How forecasts are used (service windows, expected fill level, probability of overflow)

How it helps answer RQ3:

You can identify:

- Which routing techniques are most commonly used
 - Which integrate AI forecasts and which do not
 - Whether using forecasts improves routing efficiency
 - What integration strategies exist (and where the gaps are)
-

RQ4: Digital twins

“How is digital twin technology used to model and optimize urban waste collection?”

Extract:

- Type of digital twin (monitoring, simulation, prediction, optimization)

- Data sources used (sensor data, forecasting output, routing results)
- Visualization/dashboards
- Scenario testing (weather, events, unexpected overflow)
- Use of real-time updates
- Whether it is integrated with routing or forecasting

How it helps answer RQ4:

You can compare:

- What kinds of DTs exist in the literature
- Whether they are simple dashboards or real operational twins
- Whether they support decision-making or only visualization

RQ5: Operational and environmental benefits

“What operational and environmental benefits have been demonstrated by AI- and IoT-based waste management systems?”

Extract:

- Reported performance improvements:
 - Distance saved
 - Fuel saved
 - CO₂ emissions reduced
 - Overflow incidents reduced
 - Number of trips decreased
 - Worker time saved
- KPI reporting methods (e.g. for distance saved km before and after, % of reduction)
- Study context (campus, city district, full city)
- Validation setting

How it helps answer RQ5:

You can compare:

- Actual benefits demonstrated in real deployments
- Which technologies produce measurable improvements
- Whether benefits are consistent across cities and systems

Example:

Let's say you coded 3 papers and you have this:

From RQ2 codes:

- Paper A: LSTM → RMSE 9%
- Paper B: GRU → RMSE 10%
- Paper C: XGBoost → RMSE 14%

You can now answer RQ2:

LSTM performed best across multiple studies, followed by GRU, while tree-based models performed worse.

You can answer more questions/identify more gaps by the data you will extract from the coding scheme for each RQ.